

ORIGINAL ARTICLE

Survival analysis of patients with liver cirrhosis based on deep learning to quantify body composition

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Abstract

Early body composition analysis in patients with liver cirrhosis is important for correcting malnutrition and improving prognosis and quality of life. Severe muscle wasting or Sarcopenia is the most common and often undetected complication, with serious negative effects on survival and quality of life. The purpose of this study was to use deep learning methods to segment and quantify body composition, and to determine Skeletal muscle reduced Visceral Obesity (SVO) by Skeletal Muscle Index (SMI) and Visceral to Subcutaneous adipose tissue Ratio (VSR) to examine its relationship with survival in patients with cirrhosis. In this paper, a new segmentation framework MCAUnet based on deep learning is proposed. The framework adds an attention mechanism from the perspective of the channel, which can adaptively fuse enough channel features to facilitate complex medical image segmentation. Compared with the previous segmentation model, the effect is optimal (Dice = 0.952). We analyzed 117 adult patients with cirrhosis who were admitted to Shanxi Bethune Hospital between January 2016 and December 2020. SOV was defined as Sarcopenia (male SMI < 43.24 cm² / m², female SMI < 35.11 cm² / m²) and visceral obesity (male VSR ≥ 1.07, female VSR ≥ 0.68). 18.8 % of the subjects met the SOV criteria. Through survival analysis, the 3-year and 5-year survival rates of SVO patients were significantly lower than those of normal patients (32 % VS 50 %, 5 % VS 32 %) [P = 0.035]. In regression analysis, SVO was associated with mortality in patients with cirrhosis (hazard ratio 0.54, 95 % confidence interval 0.3-0.97), and still significant after multiple regression analysis (hazard ratio 0.62, 95 % confidence interval 0.31-1.3). Overall, SVO is associated with reduced survival in patients with cirrhosis, and further prospective studies and studies in other populations are needed to evaluate the model and the actual predictive effect of SVO.

KEYWORDS

Cirrhosis, Body Composition, Skeletal muscle reduced Visceral Obesity

1 | INTRODUCTION

Liver cirrhosis is the result of the development of different forms of necrotizing liver disease, leading to liver fibrosis. Approximately 1.3 million people die annually from complications associated with liver fibrosis¹. As the main organ for maintaining metabolic homeostasis, the liver plays a pivotal role in ensuring the body's nutritional supply^{2,3}, which is involved in the absorption and metabolism of glucose, proteins, and lipids. Studies have shown that the prevalence of malnutrition in patients with cirrhosis is 20% to 50%⁴. It is also closely associated with the onset of complications such as liver encephalopathy, peritoneal effusion, Sarcopenia, and increased morbidity and mortality^{5,6}, which affects the prognosis of patients with liver cirrhosis seriously.

Although nutritional status plays an important role in the prognosis of cirrhosis, liver fibrosis is often overlooked because nutritional assessment can be complex for patients with cirrhosis who have fluid retention and are overweight^{7,8}. Child-Turcotte-Pugh⁹ score for assessing mortality in patients with cirrhosis includes the nutritional status assessment. However, most of these

Abbreviations: SVO, Skeletal muscle reduced Visceral Obesity; SMI, Skeletal Muscle Index; VSR, Visceral to Subcutaneous adipose tissue Ratio; CT, Computed Tomography.

techniques have limits because of a lack of objectivity, repeatability, and prognosis. In recent years, body composition measured by CT scans has become the gold standard and known driver of metabolic syndrome and nutritional status^{10,11,12}. Furthermore, more and more people now realize that fat and muscle mass are important determinants of patient outcomes. In other words, once a patient is affected by cirrhosis, changes in body composition can be a substitute for the nutritional status of patients with cirrhosis. However, body composition needs to be clearly defined and quantified for clinical care. In this regard, quantification of body composition with cross-sectional imaging studies [computed tomography (CT) or magnetic resonance imaging (MRI)] is an objective and reproducible technique, which is an attractive indicator of nutritional status in cirrhotic patients because it is not affected by changes in fluid status, bone density or ascites^{13,14}.

So far, there is substantial evidence that CT measurements of muscle distribution and adipose tissue have a prognostic value in patients with cirrhosis^{15,16,17,18}. However, the standard method of measuring body composition in CT scans involves tedious manual outlining^{19,20}. This takes time and is not clear due to variations between observers and between observers, therefore it is also unrealistic to apply it to clinical care²¹. The rapid development of computer technology has academic scholars to segment images from the perspective of deep learning. As the most widely used network architecture in the field of medical image segmentation, U-Net has been continuously optimized and improved by scholars. TransUNet (Chen, 2021)²² was the first Transformer-based medical image segmentation framework. Then Swin-Unet (Cao et al., 2021)²³ achieved the most advanced performance and proposed the first pure Transformer-based U-shaped architecture which introduced Swin Transformer to replace the convolution blocks in U-Net. However, the above methods mainly focus on the flaws of the convolution operation rather than on the U-Net itself, which can lead to structural redundancy and high computational costs.

In this work, we reconsider the design of skip connections in U-Net and propose the MCAUnet model to better connect features between the encoder and decoder stages. The frame adds a channel focus mechanism for sufficiently merging channel characteristics sufficient for the segmentation of complex medical images. This model was used to predict the body composition of 117 adult patients with liver cirrhosis who were admitted to Shanxi Bethune Hospital from January 2016 to December 2020. We quantified the body composition based on the segmentation results and developed a definition of SOV for the Asian population. SOV was defined as Sarcopenia (male SMI < 43.24cm² / m², female SMI < 35.11cm² / m²) and visceral obesity (male VSR ≥ 1.07, female VSR ≥ 0.68). Finally, the relationship between SOV and the outcome of patients with liver cirrhosis was analyzed.

2 | MATERIALS AND METHODS

2.1 | Study Design and Patient Choice

The training data set used in the model was composed of 11362 CT scans of the L3 lumbar spine provided by the imaging room of Bethune Hospital in Shanxi Province. The SOMATOM Definition Flash and SOMATOM Definition AS (Siemens Medical Solutions, Forchheim, Germany) CT scanner were used for the abdominal plain scan. The specific parameters were as follows. 120 kV tube voltage and 120-240 mA baseline tube current. Alignment width 0.6 × 128 mm, rotation time 0.5 s / cycle, scan layer thickness 5 mm, image reconstruction layer thickness 1 mm. Each one corresponds to a label map of subcutaneous fat, skeletal muscle, and visceral fat manually labeled by professional physicians. Both the training and the test sets used an improved and not improved CT combination, where we have already demonstrated their equivalence in cross-sectional area measurements.

The Clinical Validation Dataset used CT scans from a cohort of cirrhotic patients (n = 117) in Shanxi Bethune Hospital to study model performance and its clinical relevance. We retrospectively analyzed 117 adult patients diagnosed with cirrhosis at Shanxi Bethune Hospital from January 2016 to December 2020. In the screening process, patients with hepatocellular carcinoma or referred for hepatic transplantation were excluded. All demographic and clinical details were extracted from medical records reviews. Baseline data were obtained within 6 months of a CT scan. Deaths were confirmed by telephone follow-up and recorded in the hospital system. The last follow-up was completed in May 2022. CT scans in the clinical validation dataset were not included in the training dataset.

2.2 | Measurement of Body Composition

This article uses deep learning methods to measure body composition. As far as we know, the current segmentation methods based on Transformer mainly focus on improving the encoder of U-Net, which is based on its advantages of capturing remote information. Previous deep learning models focused primarily on defects in convolution operations, not U-Net itself. The

model MCAUnet proposed in this paper mainly solves the problem that the features of the encoder and decoder in U-Net are inconsistent. In some cases, because of the semantic difference between the superficial encoder and the decoder, superficial functionality with less semantic information can affect final performance through simple connections. Inspired by this, this paper constructs an MCAUnet framework and designs a Transformer module divided by a channel between an ordinary U-Net encoder and decoder to better integrate encoder features and reduce the semantic gap.

Specifically, we propose a channel converter (MCA) to replace the skip connection in U-Net, which combines the advantages of multi-scale encoder features and long-dependence modeling in Transformer. The framework adds an attention mechanism from the channel perspective, which can sufficiently fuse enough channel features for complex medical image segmentation. The model structure of MCAUnet are shown in Figure1.

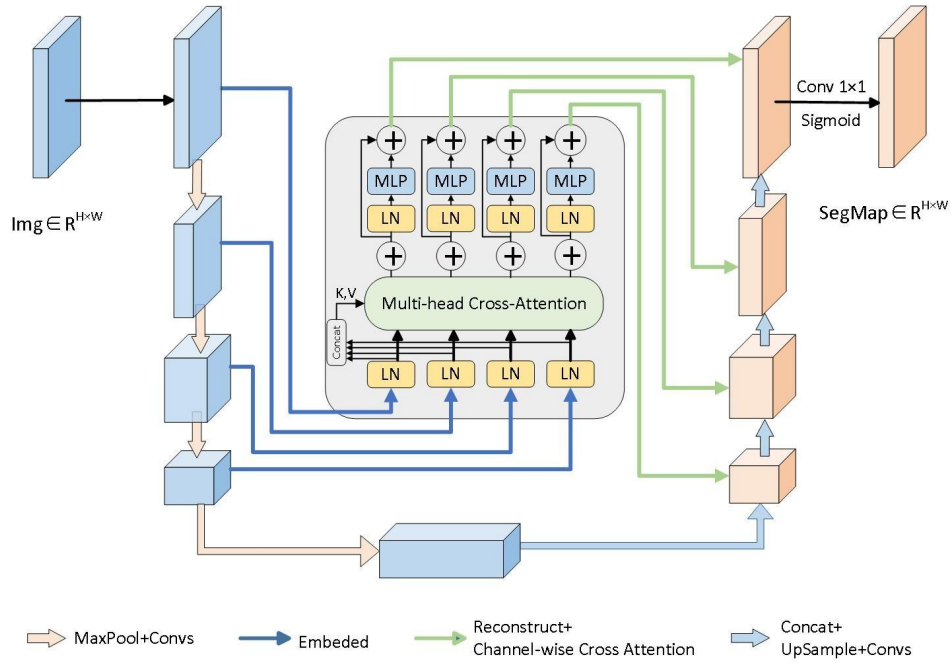


FIGURE 1 Model structure of MCAUnet

To make the input image meet the size and channel requirements of the architecture, the single channel and gray image are copied to obtain a three-channel RGB (red, green, and blue) image with a size of $512 \times 512 \times 3$. Each pixel in each image is divided into four categories: subcutaneous fat, skeletal muscle, visceral fat, and background. Training data is added to the composite data by rotating 90° , 180° , and 270° counterclockwise and flipping it with an x-axis mirror.

The total cross-sectional area (cm^2) of the corresponding body composition was found by the correspondence between the real length of the CT image and the pixels and all values were normalized with height (m^2) to obtain subcutaneous adipose tissue index (SATI, cm^2 / m^2), skeletal muscle index (SMI, cm^2 / m^2), and visceral adipose tissue index (VATI, cm^2 / m^2). VSR is obtained by dividing VATI and SATI. Due to the lack of consensus on the use of CT images to define low skeletal muscle area and low-fat area, the first quartile of the skeletal muscle or fat area index of the overall population was used as the cutoff value in this study.

2.3 | Statistical Analysis

Quantitative variables that obey the normal distribution are described by mean $\pm$ standard deviation, and quantitative variables that do not obey the normal distribution are described by median $\pm$ interquartile range. The categorical variables are described as quantity and proportion. Statistical differences were determined using the t-test with Welch correction or the Mann-Whitney U test, the Wilcoxon signed-rank test, or the Kruskal-Wallis Row test. Survival results were summarized using the Kaplan-Meier

method, and each group was compared using a log-rank test. The main results of the study were important predictors of patient death, determined using multiple Cox regression analysis proportional hazard models. Established prognostic factors for clinical death and candidate predictors of $P < 0.10$ in the regression analysis were included in multivariate Cox regression analysis. Bilateral $P < 0.05$ was considered statistically significant. The analysis was performed using SPss25.0 software (IBM, New York, USA) with RStudio version 7.2 (RStudio, New York, USA).

3 | RESULTS

3.1 | MCAUnet Model Segmentation Results

The segmentation results of the model are shown in Figure2. The artificially delineated Ground truth provides the basic facts for training the deep learning model, which is used to measure subcutaneous fat (green part), skeletal muscle (red part), and visceral fat (blue part).

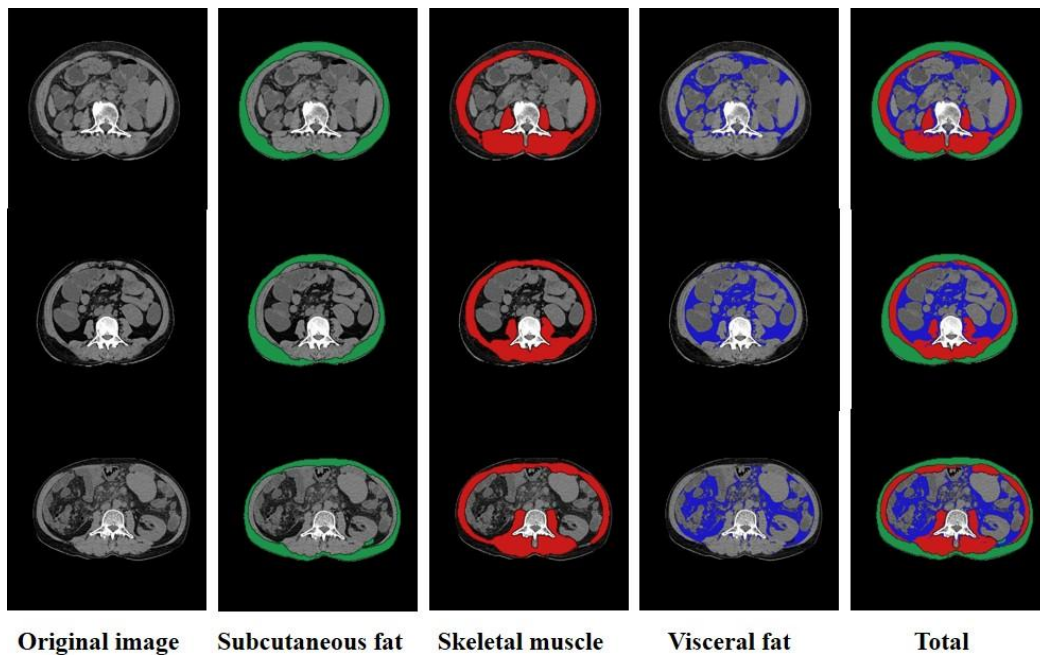


FIGURE 2 Model segmentation results

To evaluate the performance of the MCAUnet model proposed in this paper, we compare it with the other three advanced models U-Net, Pspnet, and DeeplabV3+, and evaluate the model performance across 185 clinical validation sets. The Dice coefficient and IOU (Intersection over Union) are used to evaluate the similarity between the predicted results of the model and the true values, respectively. The PPV (Positive Predictive Value) and sensitivity are used to evaluate the ability of the model to predict the correct results.

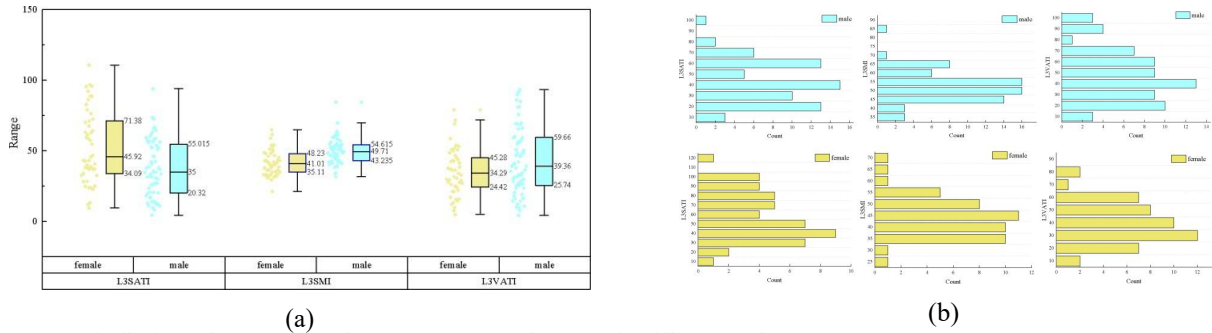
The experimental results are reported in Table1, where the best results are presented in bold. We find that the proposed model outperforms the other three advanced models in terms of all metrics. The Dice coefficient and the average boundary contour score are used to evaluate the spatial overlap effect, and the optimal values are 0.952 and 0.908, respectively. The optimal values of PPV and sensitivity were 0.952 and 0.983, respectively. This shows that the model proposed in this paper solves the limitations of the previous model, produces better segmentation results, and is more suitable for segmentation of complex medical images.

TABLE 1 Quantitative Results of Models on Test Datasets

Model	Dice	IOU	PPV	sensitivity
U-Net	0.928	0.867	0.886	0.947
PspNet	0.746	0.678	0.762	0.735
DeepLabV3+	0.906	0.830	0.891	0.922
MCAUnet	0.952	0.908	0.952	0.983

3.2 | Results of Human Body Composition

To evaluate the clinical relevance and practicability of the deep learning model, we quantified the segmentation results in the previous section and calculated the distribution of SATI, SMI, and VATI in different genders as shown in Figure3. Since the values of L3SATI, L3SMI, and L3VATI are all normal distribution variables, the first quartile is used as the cut-off value of the low subcutaneous fat index, low skeletal muscle index, and low visceral fat index in this study. The definitions of the low subcutaneous fat index, low skeletal muscle index, and low visceral fat index in different genders are as follows. L3SATI < 20.32 cm²/m² and L3SATI < 34.09 cm²/m² in both males and females had low subcutaneous fat levels. L3SMI < 43.24 cm²/m² and L3SMI < 35.11 cm²/m² in both males and females had low skeletal muscle levels. L3VATI < 25.74 cm²/m² and L3VATI < 24.42 cm²/m² L3SMI < 43.24 cm²/m² and L3SMI < 35.11 cm²/m² in both males and females had low visceral fat levels.

**FIGURE 3** Distribution and Demarcation of L3SATI, L3SMI, and L3VATI in Different Gender

3.3 | The clinicopathological features of the study population

Studies have shown that based on CT muscle and adipose tissue measurements combined to define SVO can better determine the morphological characteristics of obesity associated with the outcome of the disease. Therefore, we use the VSR to assess visceral obesity. To determine the optimal threshold for VSR to define visceral obesity, we first assessed the relationship between post-transplantation mortality and VSR based on differences in visceral and subcutaneous fat observed in our cohort. Based on the VSR quantified previously, we used the VSR cut-off point of male ≥ 1.07 and female ≥ 0.68 to define visceral obesity. SOV was defined as a combination of skeletal muscle loss (male SMI < 43.24 cm²/m², female SMI < 35.11 cm²/m²) and visceral obesity (male VSR ≥ 1.07 , female VSR ≥ 0.68).

We divided 116 patients into the data set into SVO patients and Non-SVO patients. As shown in Table2, there were 22 (18.8 %) patients with SVO and 95 (81.2 %) patients with Non-SVO. Baseline clinical characteristics of patients with and without SOV are also shown in Table2. There was no difference in sex ratio and age between the two groups. In the SVO group, BMI was significantly lower ($P = 0.001$), and body weight was significantly lighter ($P = 0.008$). At the same time, the concentrations of certain metabolic rate markers were lower, such as lower creatinine (Cre) ($P = 0.047$) and lower prothrombin time (PT) ($P = 0.044$). More importantly, the MELD score was higher in the SVO group ($P = 0.003$), which represented a higher severity of cirrhosis.

TABLE 2 Clinical demographic and laboratory characteristics of Sarcopenic visceral obesity

Variable	Overall n=117	Non-sarcopenic visceral obesity n=95	Sarcopenic visceral obesity n=22	P-value
sex = 1 (%)	68 (58.1)	56 (58.9)	12 (54.5)	0.891
Age	55.93 (11.05)	55.71 (10.91)	56.91 (11.86)	0.647
BMI	24.71 (3.59)	25.25 (3.52)	22.37 (2.92)	0.001
weight	65.50 [59.10, 77.50]	67.50 [60.10, 78.30]	62.00 [49.50, 68.20]	0.008
height	164.80 (8.75)	164.97 (8.46)	164.09 (10.10)	0.674
SMI	46.82 [40.26, 52.94]	48.96 [42.63, 53.75]	34.27 [32.45, 41.19]	<0.001
VSR	0.93 [0.70, 1.31]	0.87 [0.61, 1.25]	1.26 [1.01, 1.63]	0.001
ALT	56.88 (68.53)	60.71 (73.41)	40.30 (38.24)	0.209
AST	58.30 [35.70, 96.70]	58.30 [38.70, 83.55]	57.00 [29.68, 108.95]	0.53
ALP	117.40 [83.50, 146.80]	109.20 [84.35, 144.50]	130.90 [85.10, 192.73]	0.261
r-GT	78.80 [39.90, 163.90]	68.20 [37.50, 146.90]	123.65 [48.45, 223.52]	0.149
TBIL	32.60 [21.50, 69.50]	31.40 [21.85, 74.80]	33.05 [20.62, 60.25]	0.563
DBIL	10.90 [5.80, 25.70]	10.40 [5.80, 27.35]	12.15 [4.40, 16.75]	0.688
IBIL	20.40 [15.60, 43.90]	21.50 [16.00, 48.55]	18.65 [12.47, 36.95]	0.25
Alb	30.58 (5.78)	30.72 (5.93)	29.99 (5.18)	0.597
Cre	67.60 [57.30, 77.30]	69.20 [58.70, 77.30]	59.50 [49.65, 70.65]	0.047
BUN	4.80 [3.80, 6.10]	4.90 [3.90, 6.15]	4.05 [3.30, 5.72]	0.238
PT-S	14.90 [13.00, 16.90]	15.00 [13.30, 16.90]	13.70 [12.33, 15.38]	0.044
INR	1.37 [1.20, 1.55]	1.38 [1.23, 1.56]	1.27 [1.14, 1.42]	0.043
PTA	62.00 [51.00, 76.00]	60.00 [50.00, 72.50]	73.50 [59.00, 79.75]	0.068
WBC	3.90 [2.70, 5.40]	3.90 [2.70, 5.10]	4.20 [2.65, 5.57]	0.558
NAP	2.25 [1.48, 3.27]	2.21 [1.47, 3.18]	2.73 [1.64, 3.58]	0.345
TLC	1.07 (0.70)	1.07 (0.71)	1.04 (0.64)	0.864
RBC	3.57 (0.78)	3.61 (0.78)	3.39 (0.77)	0.228
HB	115.31 (25.81)	116.93 (25.31)	108.32 (27.39)	0.159
PLT	76.00 [54.00, 114.00]	69.00 [54.50, 112.50]	82.00 [54.00, 145.75]	0.307
CPG (%)				0.355
A	26 (22.2)	22 (23.2)	4 (18.2)	
B	48 (41.0)	36 (37.9)	12 (54.5)	
C	43 (36.8)	37 (38.9)	6 (27.3)	
CPS	9.00 [7.00, 11.00]	9.00 [7.00, 11.00]	8.00 [7.00, 9.75]	0.487
MELD	7.82 [5.12, 11.60]	4.15 [-0.69, 8.64]	8.02 [6.17, 11.99]	0.003

SMI: skeletal muscle index, VSR: Ratio of visceral to subcutaneous adipose tissue, ALT: glutamic pyruvic transaminase, AST: glutamic-oxalacetic transaminase, ALP: alkaline phosphatase, r-GT: Bile duct related enzymes, TBIL: total bilirubin, DBIL: conjugated bilirubin, IBIL: indirect bilirubin, Alb: albumin, Cre: creatinine, BUN: blood urea nitrogen, PT-S: prothrombin time, INR: international normalized ratios, PTA: prothrombin activity, WBC: white blood cell count, NAP: neutrophil counts, TLC: lymphocyte numbers, RBC: red-cell count, HB: hemoglobin, PLT: platelet count, CPG: Child-Pugh grading standard, CPS: Child-Pugh score, MELD: a model for end-stage liver disease

3.4 | Survival analysis of SVO patients

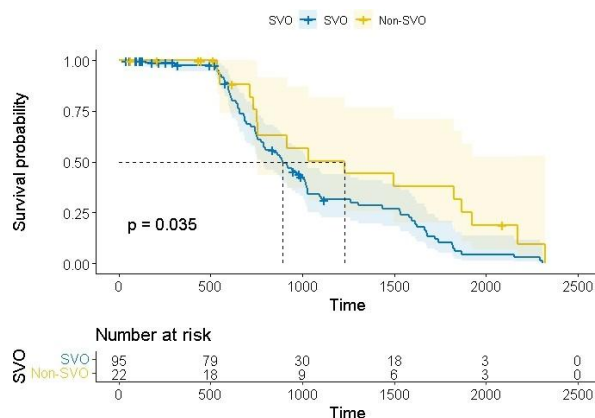
Recognition of body composition as an important determinant of mortality in patients with cirrhosis²⁴, We used deep learning models to extract body composition features from clinical CT scans to predict mortality. In Cox proportional hazard regression analysis, we found that many components of body composition were significantly associated with mortality. As shown in Table3, Cox regression analysis shows that BMI (HR 1.06, 95 % CI 1-1.1, P = 0.03), sarcopenic visceral obesity (SVO, HR 0.54, 95 % CI 0.3-0.97, P = 0.038), alkaline phosphatase (ALP, HR 0.99, 95 % CI 0.99-1, P = 0.028), total bilirubin (TBIL, HR 1.00, 95 % CI 1-1, P = 0.015), conjugated bilirubin (DBIL, HR 1.00, 95 % CI 1-1, P = 0.011), indirect bilirubin (IBIL, HR 1.00, 95 % CI 1-1, P = 0.041), prothrombin time (PT-S, HR 1.08, 95 % CI 1-1.2, P = 0.016), international normalized ratio (INR, HR 2.38, 95 % CI 1.2-4.8, P = 0.014), prothrombin activity (PTA 95 % CI 1.02), model of end-stage liver disease (MELD, HR 1.04, 95 % CI 1-1.1, p=0.013) was significantly associated with postoperative OS. In multivariate Cox regression, SVO was still independently associated with post-transplant mortality. As shown in Table3, sarcopenic visceral obesity (SVO, HR 0.62, 95 %

TABLE 3 Cox regression analysis of overall survival rate

Variable	Single Factor Analysis		Multiple-Factor Analysis	
	HR (95% CI)	P-value	HR (95% CI)	P-value
sex	1.31 (0.85-2)	0.219		
Age	0.99 (0.97-1)	0.370		
BMI	1.06 (1-1.1)	0.030		
weight	1.01 (1-1)	0.080		
height	1.00 (0.98-1)	0.758		
SMI	1.01 (0.99-1)	0.278		
VSR	1.10 (0.71-1.7)	0.667		
SVO	0.54(0.3-0.97)	0.038	0.62 (0.31-1.3)	0.046
ALT	1.00(1-1)	0.791		
AST	0.99(0.99-1)	0.235		
ALP	0.99 (0.99-1)	0.028	1 (0.99-1)	0.023
r-GT	0.99 (1-1)	0.409		
TBIL	1.00 (1-1)	0.015		
DBIL	1.00 (1-1)	0.011		
IBIL	1.01 (1-1)	0.041		
Alb	0.98(0.95-1)	0.386		
Cre	1.01 (1-1)	0.056		
BUN	1.10 (0.99-1.2)	0.090		
P-TS	1.08(1-1.2)	0.016		
INR	2.38(1.2-4.8)	0.014		
PTA	0.98(0.97-1)	0.022		
WBC	1.06 (0.98-1.2)	0.118		
NAP	1.09(0.99-1.2)	0.070		
TLC	1.06(0.72-1.6)	0.765		
RBC	0.79(0.59-1.1)	0.107		
HB	0.99(0.99-1)	0.582		
PLT	0.99(0.99-1)	0.211		
CPS	1.09(0.99-1.2)	0.089		
MELD	1.04 (1-1.1)	0.013		

CI 0.31-1.3, $P = 0.046$) and alkaline phosphatase (ALP, HR 1.00, 95 % CI 0.99-1, $P = 0.023$) were independent prognostic factors for OS in patients with cirrhosis.

Next, we performed a survival analysis of patients with SVO. During the 5-year follow-up, no patients were lost to follow-up and 28 patients died. As shown in Figure4, the 3-year cumulative survival rate (OS) of cirrhotic patients with or without SVO was 32 % and 50 %, respectively and the 5-year cumulative survival rate (OS) of cirrhotic patients with or without sarcopenic obesity was 5 % and 32 %, respectively. This indicates that the cumulative incidence of death in cirrhotic patients with sarcopenic obesity was significantly higher. The presence of SVO was significantly associated with the cumulative incidence of mortality in patients with cirrhosis ($P = 0.035$).

**FIGURE 4** Survival analysis of patients with liver cirrhosis differentiated by muscular reduction visceral obesity

3.5 | Comparison of Different Standard Methods

We evaluated the improvement in predicting postmortem mortality when using VSR to define visceral obesity. Compared with other criteria used in previous studies, we used the MELD (model for end-stage liver disease) scoring system. $MELD \leq 14$ is defined as low-risk, and $MELD > 14$ is defined as intermediate-risk or high-risk patients. It can be seen that there is no statistically significant relationship with the survival rate of patients with cirrhosis as shown in Figure5.

In addition, if the patients with SVO were defined by the sum of fat area (total fat area index(TFAI) $\geq 46.06\text{cm}^2/\text{m}^2$ in males, $TFAI \geq 58.51\text{cm}^2/\text{m}^2$ in females, $SMI < 43.24\text{cm}^2/\text{m}^2$ in males , $SMI < 35.11\text{cm}^2/\text{m}^2$ in females) , it can be seen that there is no statistically significant relationship between the survival rate of patients with cirrhosis as shown in Figure6. It shows that the use of SVO to distinguish patients to predict the overall survival of patients with cirrhosis has high reliability.

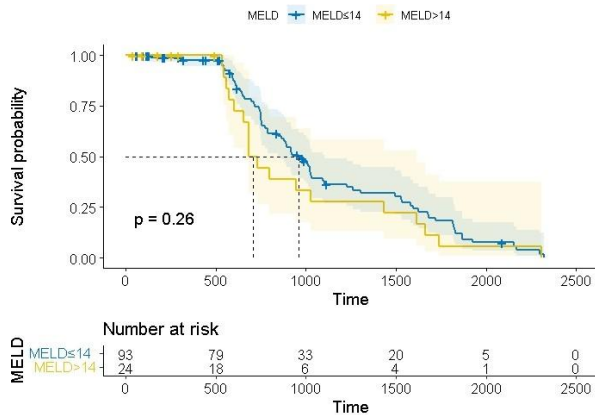


FIGURE 5 Survival analysis of patients with liver cirrhosis by MELD score

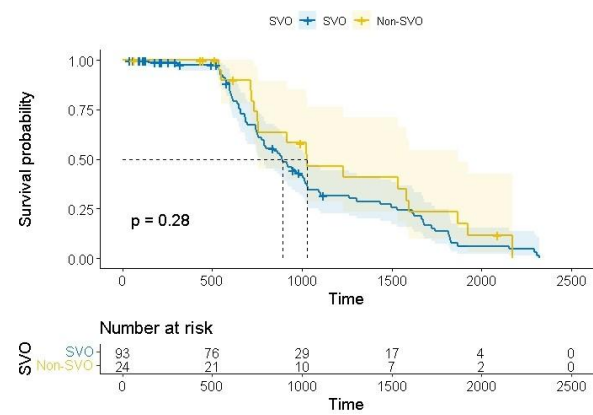


FIGURE 6 Survival analysis of patients with liver cirrhosis based on total fat content

4 | DISCUSSION

In this study, we proposed an MCAUnet model based on deep learning for body composition segmentation. The model reconsidered the design of skip connections in U-Net to better connect the features between the encoder and decoder stages. The framework added an attention mechanism from the channel perspective, which can adaptively fuse multi-channel features for complex medical image segmentation. We used this model to perform composition prediction in 117 adult cirrhotic patients attending Baijuan Hospital in Shanxi between January 2016 and December 2020, and quantified body composition based on segmentation results to develop Asian-compliant SOV definition criteria. The SOV definition criteria in line with Asians was defined as sarcopenia (male $SMI < 43.24\text{cm}^2/\text{m}^2$, female $SMI < 35.11\text{cm}^2/\text{m}^2$) and visceral obesity (male $VSR \geq 1.07$, female $VSR \geq 0.68$). Then we used Cox proportional hazard regression model to analyze the risk factors of mortality in patients with cirrhosis and proved the correlation between SOV and adverse outcomes in patients with cirrhosis. Finally, we tried to use the MELD score and other definitions of visceral obesity to verify the accuracy of SOV.

In previous studies, the majority of patients with cirrhosis had complications such as fluid retention, resulting in large discrepancies between body composition results based on conventional measures and the true results. Because in most of these patients, body composition is likely masked by fluid retention. To achieve more accurate body composition measurements, cross-sectional CT imaging of the patient's third lumbar vertebra is becoming the primary basis for body composition measurements, especially for muscle and fat content^{25,26}. This can be of particular clinical significance for patients with chronic diseases. Because cirrhosis is a chronic wasting disease, the impact on the patient's physical function and nutritional status cannot be ignored. However, the Child-Pugh classification and the end-stage liver disease model, which have been widely used in previous studies to evaluate the reserve functions of the patient's liver, including synthesis, detoxification, and biotransformation, lacked an assessment of the patient's nutritional status. Body composition indicators such as skeletal muscle, visceral fat, and

subcutaneous fat are often used in clinical research because of their objectivity, repeatability, and quantifiable advantages in evaluating the nutritional and metabolic status of the body. However, standard methods for measuring body composition included visual inspection and time-consuming manual delineation, which were not suitable for clinical implementation. In this study, we proposed a new deep learning model MCAUnet, which can objectively measure body composition in CT scans. More importantly, this process can be fully automated using deep learning methods. We demonstrated that our convolutional neural network can accurately identify key areas of measurement and show a high correlation with manual measurements.

Through automated measurement, we quantified the content of subcutaneous fat, skeletal muscle, and visceral fat in patients with cirrhosis, and defined the critical value. In this study, we observed that nearly half of patients with liver cirrhosis had sarcopenia or visceral obesity, which was determined by CT-quantified body composition. 18.8 % of them were classified as SVO due to both sarcopenia and visceral obesity. We further observed that the presence of SVO was independently associated with lower survival rates compared to patients with non-sarcopenia and visceral obesity, resulting in 3-year and 5-year survival rates of 32 % and 5 %, respectively.

While the exact pathogenesis of the gender difference between sarcopenia and survival remains unclear, our findings raise the possibility that SVO is not simple sarcopenia but maybe a more appropriate indicator for clinical practice to identify patients with decompensated cirrhosis at increased risk of adverse events after the illness. We also used deep learning methods to quantify body composition. This new index that combined muscle and fat quantification promoted our understanding of the practical risk assessment of this population.

Furthermore, it is known that body fat distribution may be more important than overall fat by analysing the direct and inverse relationship between VAT and SAT, and the relationship with poor clinical outcomes in cirrhosis. Previous studies used CT-quantified VAT area ≥ 100 cm² to define visceral obesity in cirrhotic patients with osteoporosis, although they did not adjust for sex and/or body mass. In addition, only SVO defined by VSR was significantly correlated with cumulative survival at 36 months after transplantation, whereas no significant correlation was observed with survival when SVO was defined by other criteria used in previous studies^{27,28,29}.

The study found that a reduction in skeletal muscle was strongly associated with the incidence of complications and mortality in cirrhosis³⁰. It is also more likely to lead to prolonged hospital stay and increased hospital costs³¹. At the same time, reduced muscle content and rapid depletion lead to reduced ability to remove blood ammonia, thereby increasing the risk of hepatic encephalopathy³². Visceral obesity represents a decrease in subcutaneous fat and an increase in visceral fat that leads to muscle fat accumulation, insulin resistance, and protein metabolism disorders. It also increases the likelihood of liver cancer before and after liver transplantation³³. Therefore, the abnormal body composition indicators measured in this study are closely related to the mortality rate, liver cancer incidence, and adverse outcomes of patients with cirrhosis, and are of great help to the designation of subsequent targeted treatment strategies.

Our study also has some limits. Firstly, as we only screened the study population at the Baiqiuen Hospital in Shanxi, we propose an objective evaluation metric to assess sarcopenia versus visceral obesity and demonstrate the importance of this evaluation metric in predicting survival in patients with cirrhosis, but given our relatively small sample may not reflect the broader population of patients with cirrhosis. However, the incidence of sarcopenia, visceral obesity, and SVO in our population is similar to previous studies. These studies included more patients with different types of cirrhosis, using traditional CT to quantify body composition. In addition, the median SMI in our cohort was similar to that reported in another study specifically targeting Chinese (43.23 vs 44.77 cm² / m²)³⁴. This also proves the applicability of our research results. Secondly, another limitation is that the feasibility and acceptability of the deep learning model MCAUnet proposed in this paper have not been further studied in clinical practice. Due to the single CT imaging equipment used in the training model, the model does not have good generalization. The accuracy of the model may be reduced when applied to other data sets. Finally, small-scale research cohorts limit our ability to significantly assess the association between SVO and other disease outcomes. In addition to survival rates, larger studies are needed to verify our proposed VSR boundary in defining visceral obesity and its association with suboptimal clinical outcomes caused by sarcopenia.

In summary, our data show that using deep learning methods to quantify CT body composition to measure the relative distribution of muscle and adipose tissue can more comprehensively and objectively predict the cumulative survival rate of patients with cirrhosis. Importantly, the simultaneous presence of sarcopenia and visceral obesity was associated with the worst survival rates after the illness, rather than any of the conditions occurring alone. Our data provide a tool for the diagnosis of liver cirrhosis to assess the body composition of women and men with liver cirrhosis. Clinicians should pay enough attention to improve the awareness of preoperative diagnosis and take corresponding interventions to improve the prognosis of patients.

5 | CONCLUSION

In summary, this study shows that the presence of L3 layer SVO determined by CT images of patients with cirrhosis can be used to predict OS in patients with cirrhosis. Furthermore, compared with other criteria used in previous studies (MELD score and total body fat content), the use of SVO to distinguish patients has higher reliability in predicting the overall survival of patients with cirrhosis. Surgeons should therefore pay more attention to the presence of patients with sarcopenic visceral obesity, which can help to personalize nutritional therapy promptly, reduce postoperative complications and improve the long-term prognosis of patients.

AUTHOR CONTRIBUTIONS

W.J contributed to the methodology, software, and drafted the manuscript. W.R and Z.Y contributed to drafted the Review and Editing, Z.C and E.L contributed to the interpretation of the data and critically revised the manuscript. P.M and Z.W responsible for overseeing and editing. All authors gave final approval and agree to be accountable for all aspects of the work ensuring integrity and accuracy.

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FINANCIAL DISCLOSURE

None reported.

ADDITIONAL INFORMATION

Conflict of interest The authors declared no potential conflicts of interest with respect to the research, authorship and publication of this article.

Ethical Approval Ethics approval was granted by the Shanxi Bethune Hospital (Shanxi Academy of Medical Sciences) Medical Ethics Committee (approval number: YXLL-2022-094)

Informed Consent Informed consent was obtained from all individual participants included in the study.

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